

Figure 2: Riak CS architecture

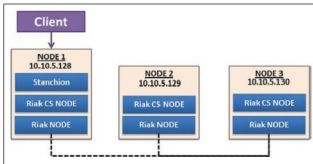


Figure 3: Riak CS cluster setup

Step 4: The chunks are then replicated to other Riak nodes within the cluster, thus providing fault tolerance and high availability.

A guide on how to build your own scalable cloud storage with Riak CS

Installing Riak CS is a fairly straightforward process involving a couple of simple steps. You can install Riak CS on most popular Linux distributions such as Red Hat, CentOS and Ubuntu. In this tutorial, I'll be setting up Riak CS on a cluster of three Ubuntu 12.04 servers as shown in Figure 3.



Note: The following steps are not recommended for running Riak CS on a production environment. These are only for testing purposes.

Prerequisites

You need to increase your Linux system's open file size limit for Riak CS to work on all nodes. Riak CS consumes a very large number of open files during its operation. To do so, edit the following file as shown:

```
# sudo vi /etc/security/limits.conf
root soft nofile 65536
root hard nofile 65536
```



Figure 4: An overview of Riak Control

```
riak soft nofile 65536
riak hard nofile 65536
```

You will also need to install a few dependencies into your system before you proceed:

```
# sudo apt-get install git build-essential libcurses5-dev
openssl libssl-dev libssl0.9.8
# sudo apt-get install erlang-base erlang-dev erlang-
parsetools erlang-eunit erlang-reltool erlang-os-mon
```

Installation

Once the prerequisites are completed, add repository keys and set up your own basho.list repository:

```
# curl -k -o- http://apt.basho.com/gpg/basho.apk
key">http://apt.basho.com/gpg/basho.apk.key</a> | sudo apt-
key add -
# bash -c "echo deb <a href='http://apt.basho.com'>http://
apt.basho.com</a> $(lsb_release -sc) main > /etc/apt/sources.
list.d/basho.list"
# apt-get update
# sudo apt-get update
```

You are now ready to install Riak CS on your local system. For Riak CS to work, you will also need to install Riak to work with it.

```
# sudo apt-get install riak riak-cs
```

Testing

Once the installation completes, you can start your Riak CS node by executing the 'riak start' and the 'riak-cs start' command as shown. If all goes well, you will not be shown any output of the command. You can check whether your Riak CS node is running by executing the 'riak ping' and also the 'riak-cs ping' command. You should get the response 'pong' if your Riak CS node has indeed started,